

Xinbao Qiao

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EDUCATION

- **The Chinese University of Hong Kong** 2026 - present
PhD student in Information Engineering Hong Kong
- **Zhejiang University** 09/2022 - 12/2025
Master in Artificial Intelligence Zhejiang
 - Relevant courses: AI Algorithms and Systems (100), Secure Artificial Intelligence (97)
 - Major GPA: 90/100 (Rank: 3/25)
- **Shandong University** 09/2018 - 07/2022
Bachelor in Communication Engineering Shandong
 - Major GPA: 82.47/100; Award: Third-class Academic Award 2018-2019

RESEARCH INTERESTS

- Lifecycle management of data in AI models, focusing on theoretical methods and practical problems that arise as data are generated, used, and deleted.
- The intersection of AI with network, including AI for communication, communication for AI.

RESEARCH EXPERIENCE

- **Research on Data-Centric ML Systems** *M.Eng. Student, 03/2023 - 12/2025*
Advisor: Prof. Meng Zhang, Zhejiang University
 - **Data Influence Attribution.** Developed theoretically guaranteed approaches to quantify and analyze the influence of data on learning models, and designed unlearning techniques to remove these influences.
 - **Trade-offs in Trustworthy Artificial Intelligence.** Investigated and developed data-centric methodologies to balance fairness, robustness, privacy, and utility in machine learning systems.
 - **Cloud-Edge Collaborative Human-Space Healthcare.** Architected cloud-edge collaborative frameworks for real-time multimodal data monitoring of human health using video, audio, and sensor signals.
- **Research on Trustworthy LLM systems** *Full-time Research Intern, 06/2025 - 12/2025*
Advisor: PANG Yan, James, National University of Singapore
 - **Illusory Pattern Perception in LLMs.** Examined whether LLMs exhibit illusory pattern perception, a known human cognitive bias, and offered an interpretable framework to account for LLM behavior.
 - **Synthetic Data Evaluation.** Developed a distributed Wasserstein collaborative algorithm, providing a private Wasserstein methodology for synthetic data evaluation when real data is limited.

SKILLS

- **Programming:** Python, PyTorch, TensorFlow, MATLAB, C/C++, SQL, Linux, Docker, Git, and $\text{\LaTeX}$.
- **Language:** Mandarin Chinese (native), English

[1] **When Sample Selection Bias Precipitates Model Collapse**

Xinbao Qiao, Xianglong Du, Wei Liu, Jingqi Zhang, Peihua Mai, Meng Zhang, Yan Pang.
Forty-Third International Conference on Machine Learning, ICML, 2026.

[2] **Beyond Binary Erasure: Soft-Weighted Unlearning for Fairness and Robustness**

Xinbao Qiao, Ningning Ding, Yushi Cheng, Meng Zhang.
Fortieth AAAI Conference on Artificial Intelligence, AAAI, 2026.

[3] **Hessian-Free Online Certified Unlearning**

Xinbao Qiao, Meng Zhang, Ming Tang, Ermin Wei.
Thirteenth International Conference on Learning Representations, ICLR, 2025.

[4] **DynFrs: An Efficient Framework for Machine Unlearning in Random Forest**

Shurong Wang, Zhuoyang Shen, **Xinbao Qiao**, Tongning Zhang, Meng Zhang.
Thirteenth International Conference on Learning Representations, ICLR, 2025.

[5] **Federated Learning as Optimal Transport: Barycentric Multi-Prototype Classification for Pretrained Model**

Xinbao Qiao, Wenjing Yan, Ying-Jun Angela Zhang.
Under review.

[6] **Illusory Pattern Perception Drives Spurious Inference in Large Language Models**

Peihua Mai, Zhuoyan Shao, **Xinbao Qiao**, Meng Zhang, Xinyue Zhou, Yan Pang.
Under review.

[7] **Learn What Matters: Data Pruning for Efficient Decentralized Learning**

Xinbao Qiao, Xunhao Jiang, Zuozhu Liu, Peng Sun, Meng Zhang.
Under review.